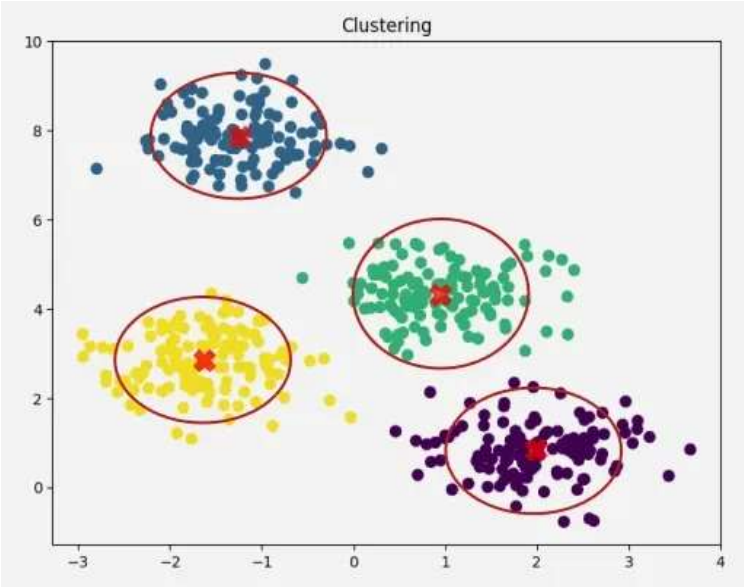
Day – 13 Clustering

Clustering is an unsupervised machine learning technique used to group similar data points together without using labelled data. It helps discover hidden patterns or natural groupings in datasets by placing similar data points into the same cluster.

- Discover the natural grouping or structure in unlabelled data without predefined categories.
- Data points are assigned to clusters based on similarity or distance measures.
- Uses Euclidean distance, cosine similarity or other metrics depending on data type and clustering method.



K means Clustering – Introduction

K-Means Clustering groups similar data points into clusters without needing labeled data. It is used to uncover hidden patterns when the goal is to organize data based on similarity.

- Helps identify natural groupings in unlabeled datasets.
- Works by grouping points based on distance to cluster centers.
- Commonly used in customer segmentation, image compression and pattern discovery.
- Useful when you need structure from raw, unorganized data.

Working of K-Means Clustering

Suppose we are given a data set of items with certain features and values for these features like a vector. The task is to categorize those items into groups. To achieve this we will use the K-means algorithm. "k" represents the number of groups or clusters we want to classify our items into.

The algorithm will categorize the items into "k" groups or clusters of similarity. To calculate that similarity we will use the Euclidean distance as a measurement. The algorithm works as follows:

1. **Initialization:** We begin by randomly selecting k cluster centroids.
2. **Assignment Step:** Each data point is assigned to the nearest centroid, forming clusters.
3. **Update Step:** After the assignment, we recalculate the centroid of each cluster by averaging the points within it.
4. **Repeat:** This process repeats until the centroids no longer change or the maximum number of iterations is reached.

The goal is to partition the dataset into k clusters such that data points within each cluster are more similar to each other than to those in other clusters.

Implementation:

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import StandardScaler

from sklearn.cluster import KMeans

df = pd.read_csv('/content/heart.csv')

X = df.drop(columns=['target'])

X_train, X_test = train_test_split(X, test_size=0.2, random_state=42)

scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)

X_test_scaled = scaler.transform(X_test)  # Transform test data using training mean & std

kmeans = KMeans(n_clusters=2, random_state=42, n_init=10)

kmeans.fit(X_train_scaled)

test_cluster_labels = kmeans.predict(X_test_scaled)

print("Test Set Cluster Assignments:")

print(pd.Series(test_cluster_labels).value_counts())
```

```
Test Set Cluster Assignments:
1    103
0     70
2     32
Name: count, dtype: int64
```

```
import matplotlib.pyplot as plt

import pandas as pd

cluster_counts = pd.Series({0: 129, 1: 76})
```

```
plt.figure(figsize=(6, 4))
bars = plt.bar(
    cluster_counts.index.astype(str),
    cluster_counts.values,
    color=['#2b5c8f', '#d95f02'],
    width=0.5,
)
for bar in bars:
    yval = bar.get_height()
    plt.text(
        bar.get_x() + bar.get_width() / 2,
        yval + 2,
        f'{int(yval)}',
        ha='center',
        va='bottom',
        fontweight='bold',
    )
plt.title('Test Set Cluster Assignments')
plt.xlabel('Cluster Label')
plt.ylabel('Number of Patients')
plt.ylim(0, 150)
plt.grid(axis='y', linestyle='--', alpha=0.7)
plt.tight_layout()
plt.show()
```

